INS3441
HIGHWAY ENGINEERING

Prof. Halit ÖZEN, PhD.

COURSE SYLLABUS

Course Objectives	Giving fundamental design knowledge for highway engineering.
Course Content	Introduction, definitions of highway components, characteristics of road users and vehicles, vehicle movements, general characteristics of highway traffic, highway capacity, selection of highway geometric standards, location survey, horizontal curves and super-elevation, transition curves, road profile and vertical curves, planning of urban roads, intersections, highways drainage.
Recommended Or Required Reading	•Highway Engineering, Nadir Yayla,2002. •Highway Constructions Design and Application, İlhan Sütaş, Güven Öztaş, 1986 •Related research papers and web sites. •A Policy on Geometric Design of Highways and Streets, AASHTO, 2001. •Highway Capacity Manual (HCM 2000), Transportation Research Board, 2000. •Highway Design and Traffic Safety Engineering Handbook, R. Lamm, B. Psarianos, T. Mailaender, McGraw-Hill, 1999. •Introduction to Transportation Engineering, James K. Banks, McGraw-Hill, 2002. •Introduction to Highway Engineering (Lecture note), Dr. Mustafa Sinan Yardım, 2012.

COURSE SYLLABUS

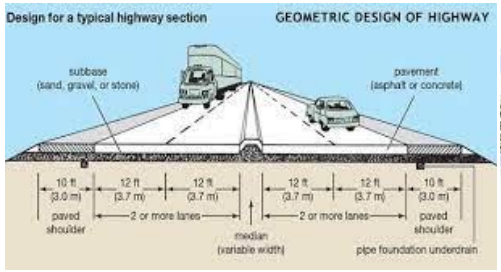
Weekly Subjects and Related Preparation Studies

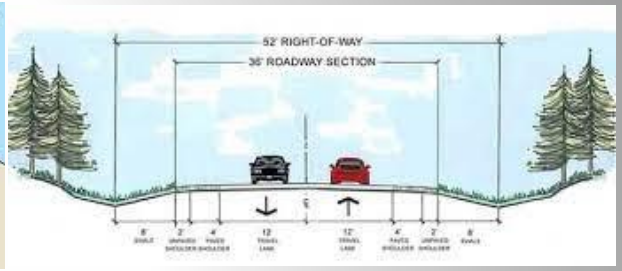
Week	Subject	Date	
1	INTRODUCTION	10/4/2021	Monday
	BASIC DEFINITIONS (TERMS)	10/6/2021	Wednesday
2	ROAD USER CHARECTERISTICS	10/11/2021	Monday
	VEHICLE MOVEMENTS	10/13/2021	Wednesday
3	TRAFFIC FLOW AND CAPACITY	10/18/2021	Monday
	ROADWAY CLASSIFICATION AND GEOMETRIC STANDARDS	10/20/2021	Wednesday
4	HIGHWAY ROUTE STUDY	10/25/2021	Monday
	HORIZANTAL CURVE AND SUPERELEVATION	10/27/2021	Wednesday
5	HORIZANTAL CURVE AND SUPERELEVATION	11/1/2021	Monday
	Project sheet distribution)-APPLICATION (VEHICLE MOVEMENTS)	11/3/2021	Wednesday
6	**ASSIGNMENT - APPLICATION (TRAFFIC)**	11/8/2021	Monday
	APPLICATION (HORIZANTAL CURVE)	11/10/2021	Wednesday
7	**ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)**	11/15/2021	Monday
	TRANSITION CURVES- ASSIGNMENT (HAND IN)	11/17/2021	Wednesday
8	**MID TERM EXAM**	11/22/2021	Monday
		11/24/2021	Wednesday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN)	11/29/2021	Monday
	TRANSITION CURVES – ASSIGNMENT (RETURN)	12/1/2021	Wednesday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE	12/6/2021	Monday
	APPLICATION (TRANSITION CURVES)	12/8/2021	Wednesday
11	**ASSIGNMENT**	12/13/2021	Monday
	APPLICATION (VERTICAL CURVE)	12/15/2021	Wednesday
12	**APPLICATION (VERTICAL CURVE)**	12/20/2021	Monday
	APPLICATION (VERTICAL AND HORIZANTAL CURVE)	12/22/2021	Wednesday
13	PLANNING OF INTERSECTION AND URBAN ROADS - PROJECTS DELIVERY DUE DATE (December 27 2020)	12/27/2021	Monday
	HIGHWAY PAVEMENT	12/29/2021	Wednesday
14	DRAINAGE IN HIGHWAY	1/3/2022	Monday
	MAKEUP EXAM	1/5/2022	Wednesday

There are three main features of roadway geometry that are decided as a part of the design.

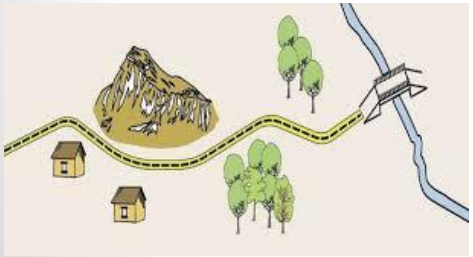
- **Cross Section**







- **Alignment**



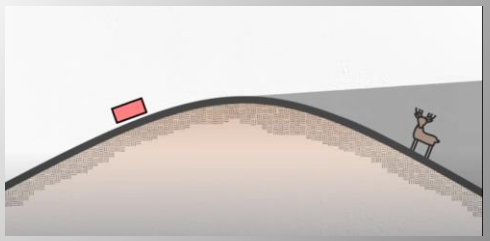




- **Profile**

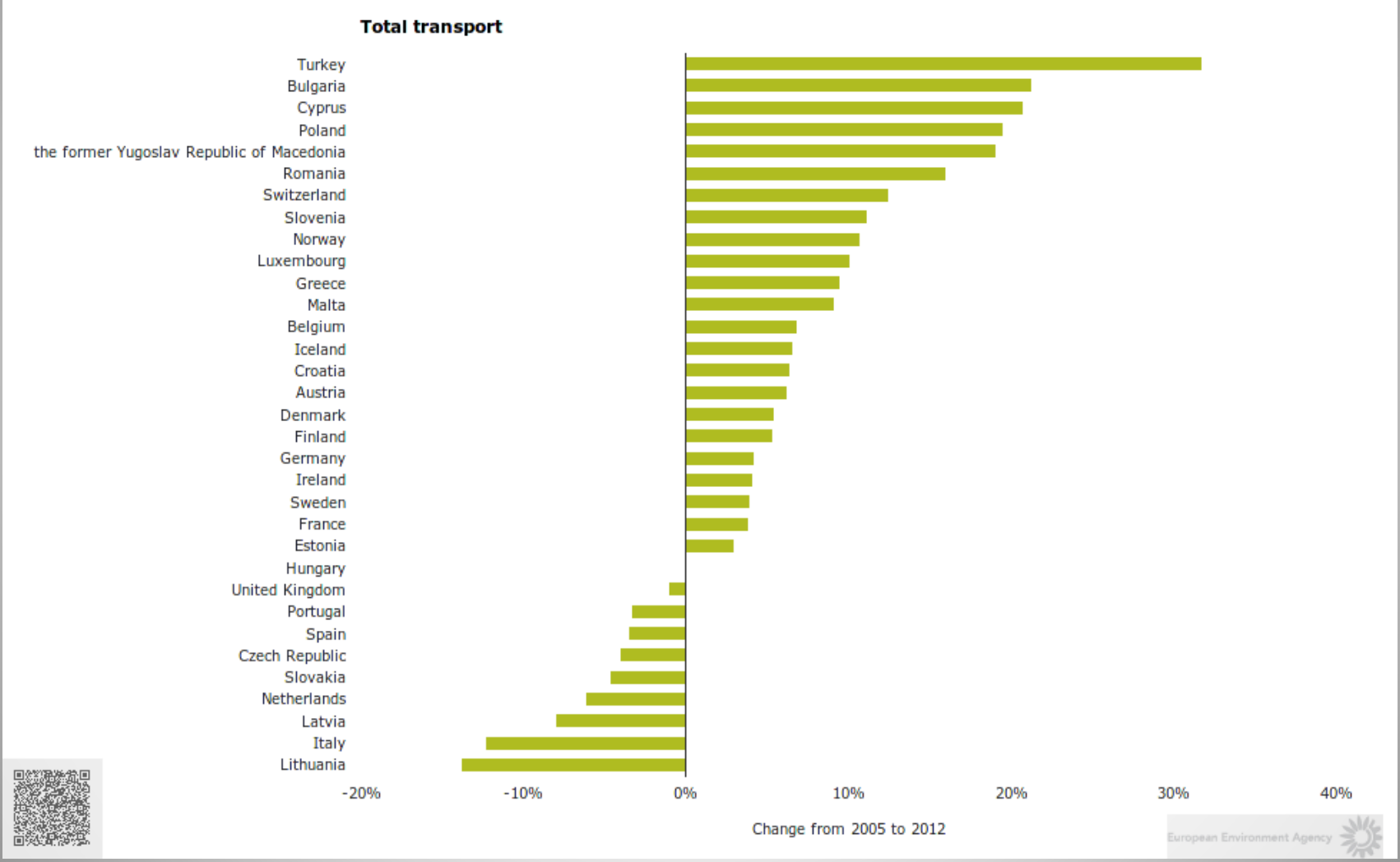






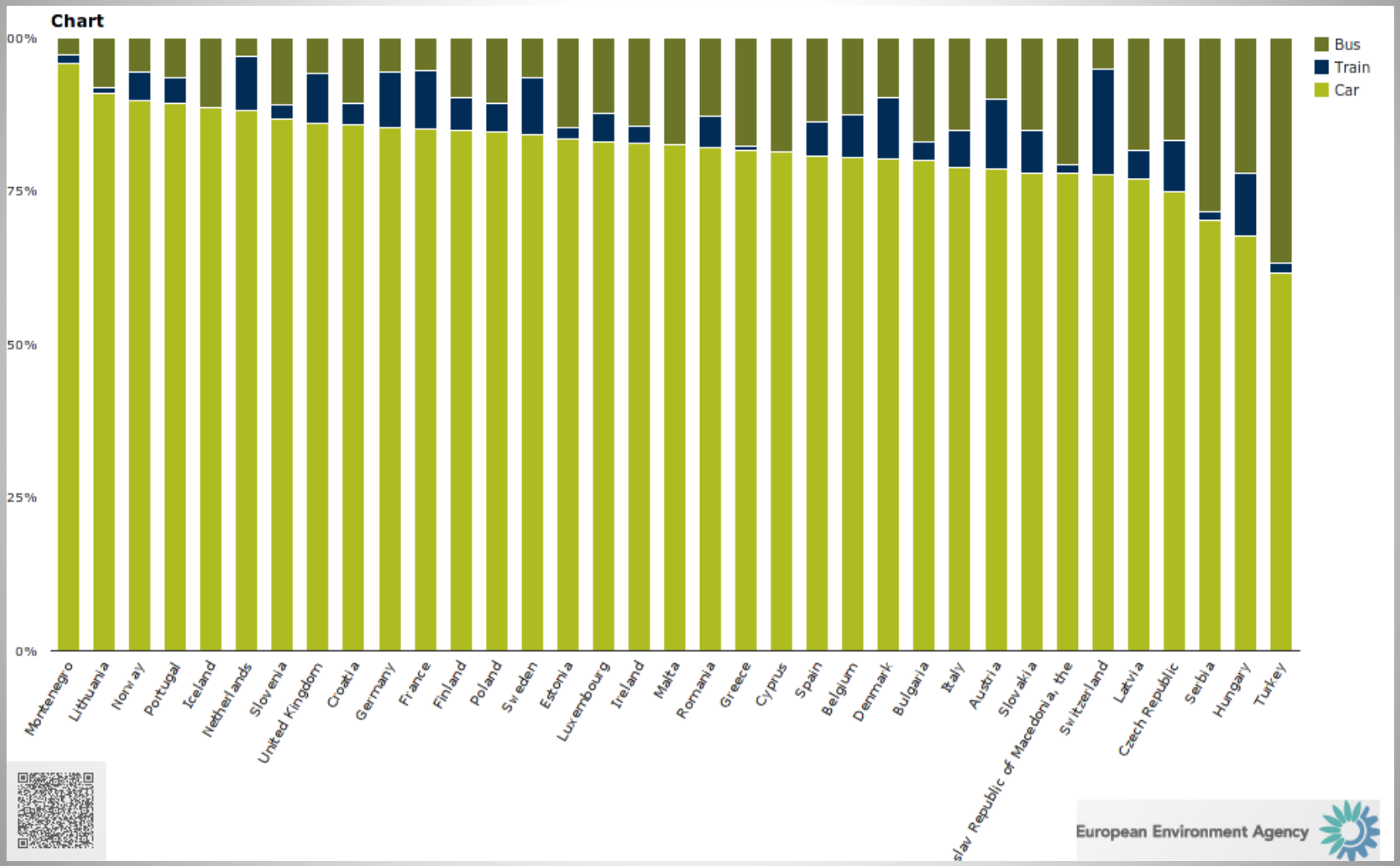
Transportation Statistics

Percentage change in total passenger transport demand in 33 European countries (2005 to 2012)



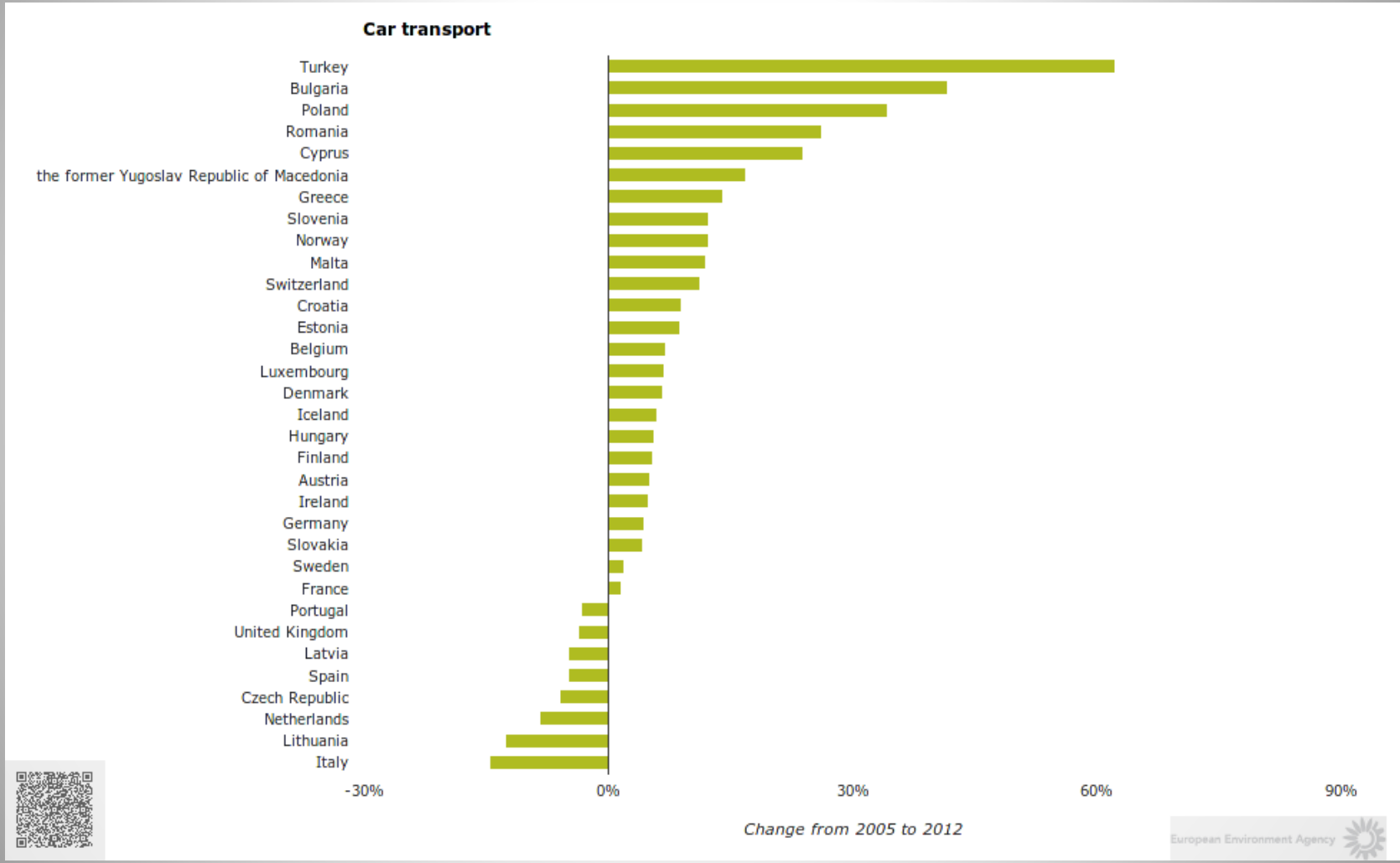
Transportation Statistics

Modal split of passenger transport in 35 European countries in 2012



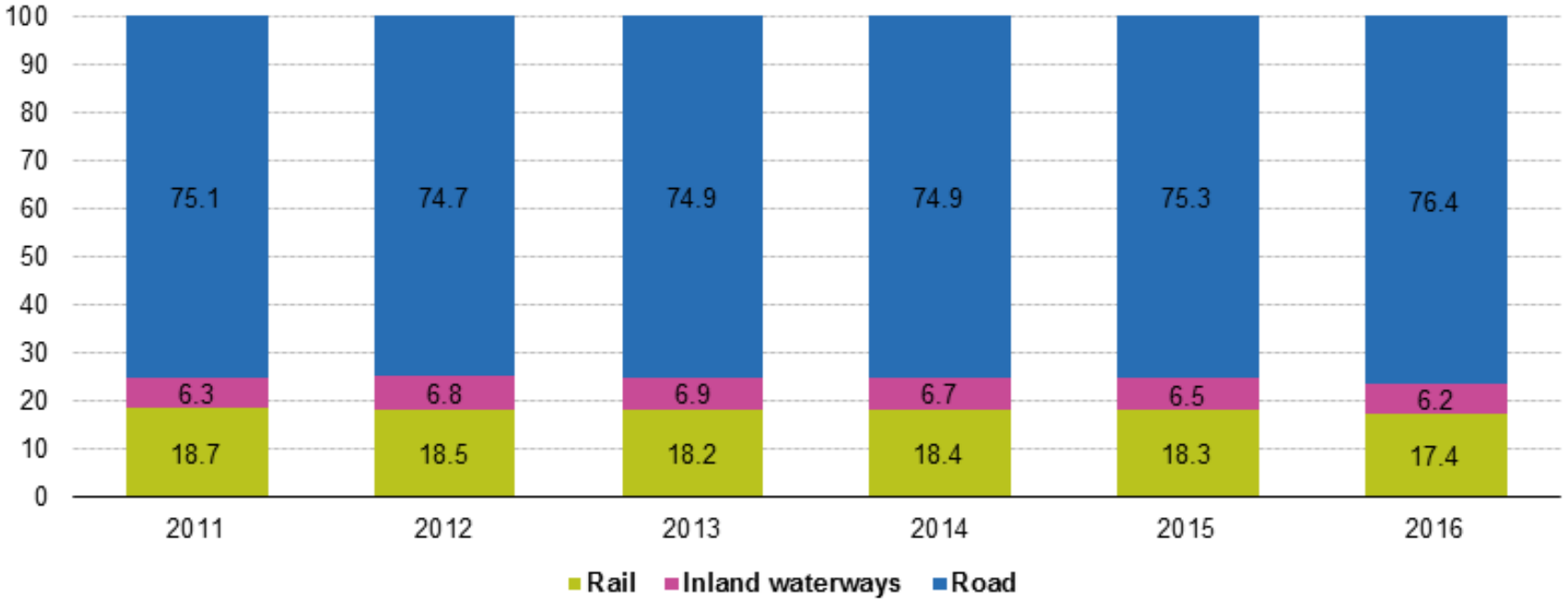
Transportation Statistics

Percentage change in car passenger transport demand in 33 European countries (2005 to 2012)



Transportation Statistics

Freight transport in the EU-28: modal split of inland transport modes (% of total tonne-kilometres)

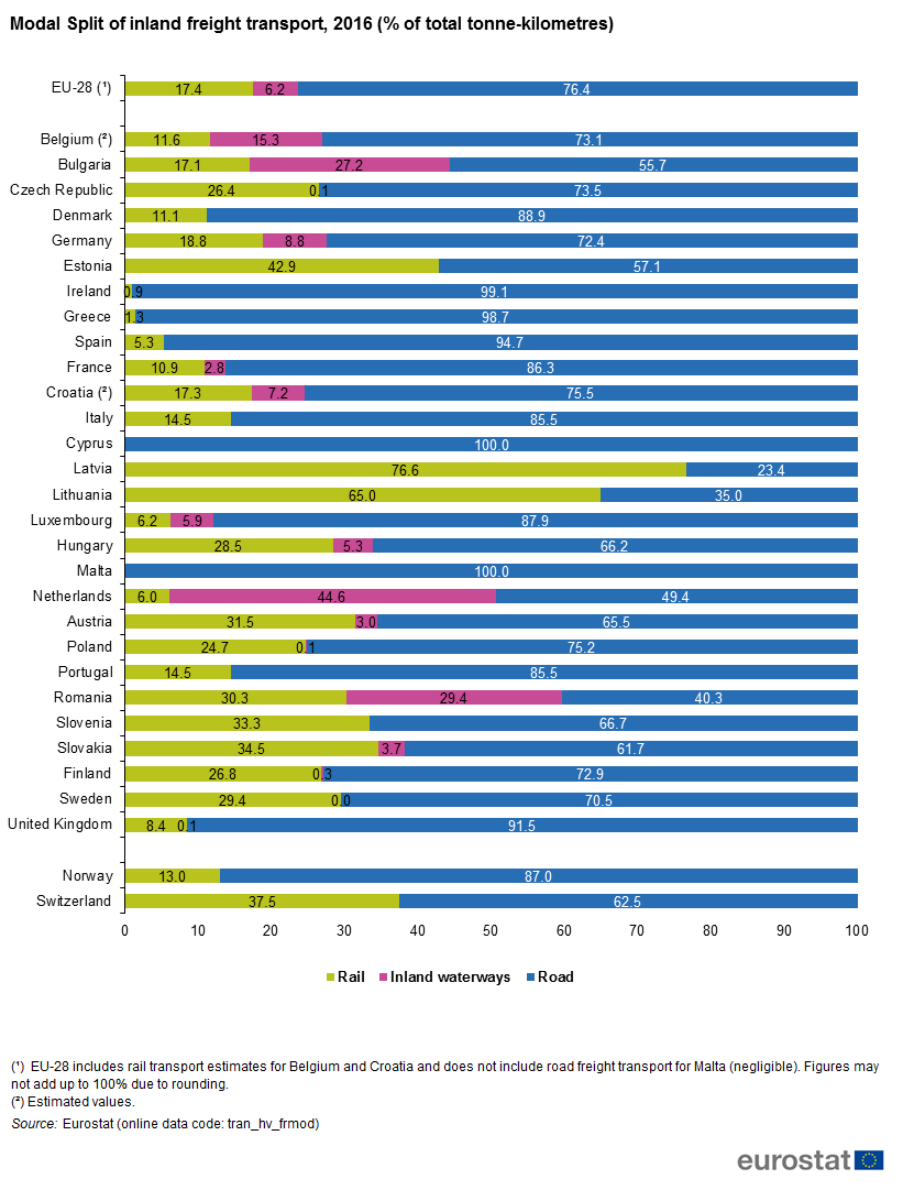


Note: EU-28 includes rail transport estimates for Belgium and Croatia and does not include road freight transport for Malta (negligible). Figures may not add up to 100% due to rounding.

Source: Eurostat (online data code: tran_hv_frmod)

eurostat

Transportation Statistics

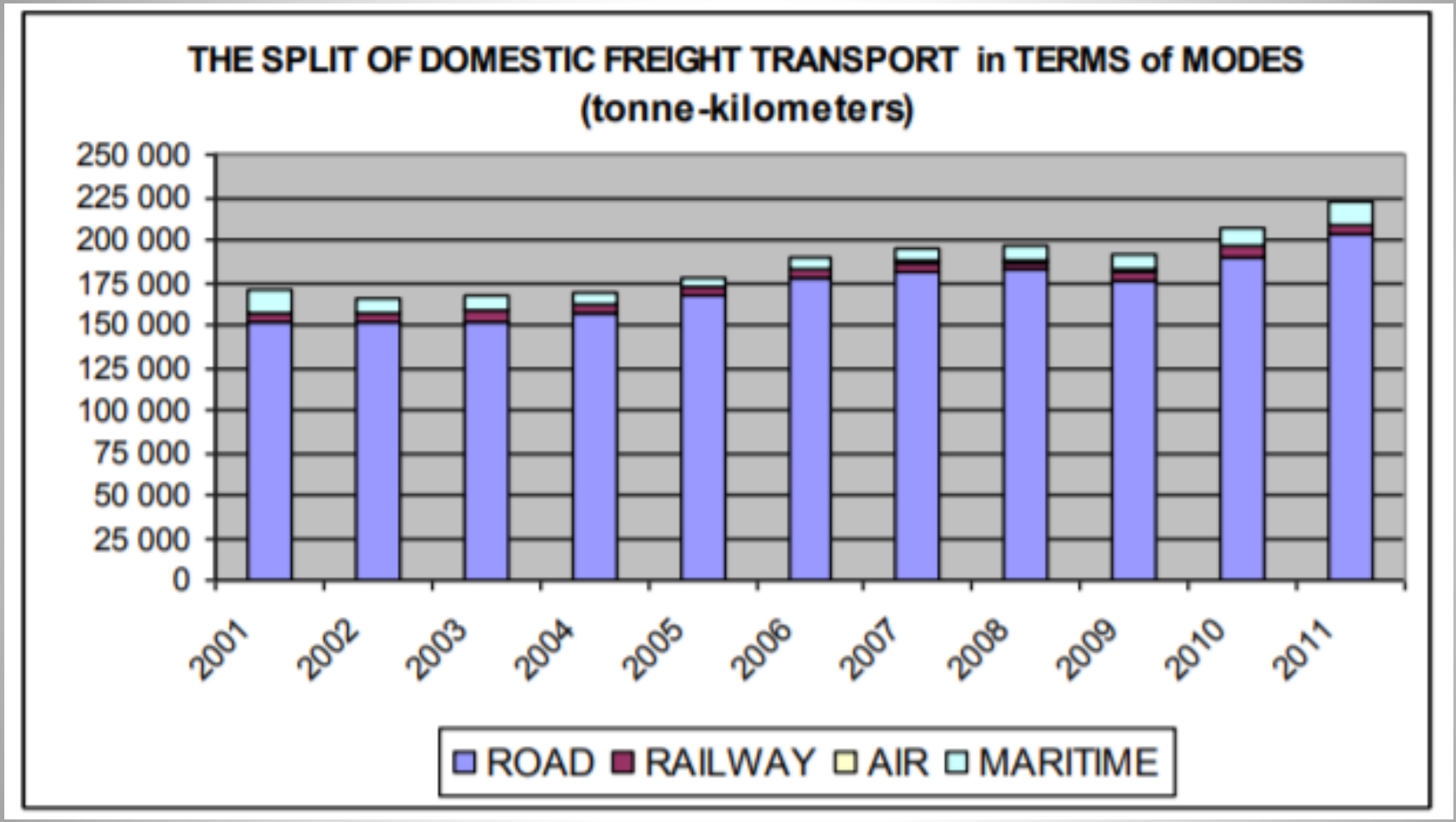


(*) EU-28 includes rail transport estimates for Belgium and Croatia and does not include road freight transport for Malta (negligible). Figures may not add up to 100% due to rounding.
(²) Estimated values.

Source: Eurostat (online data code: tran_hv_frmod)

eurostat

Transportation Statistics



Transportation Statistics

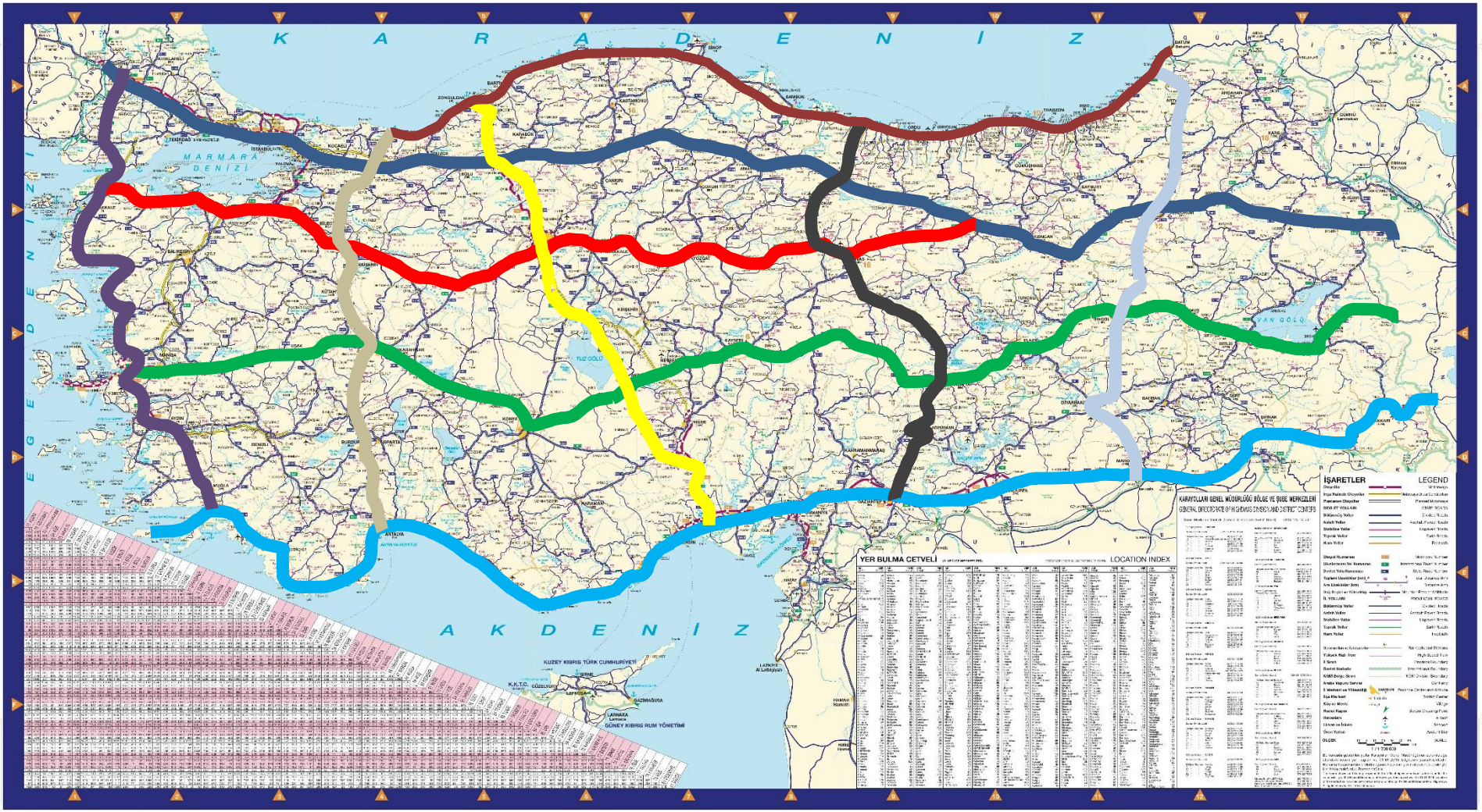
Road Network According To Surface Types (Km.)

(01.01.2018)

YOL SINIFI Road Types	ASFALT BETONU Asphaltic Concrete	SATHİ KAPLAMA Surface Treatment	PARKE Stone Block	STABİLİZE Stabilized	TOPRAK Earth	GEÇİT VERMEZ Primitive	TOPLAM UZUNLUK Total Length
OTOYOLLAR(*) Motorways	2 157	--	--	--	--	--	2 157
DEVLET YOLLARI State Highways	16 991	13 691	66	36	--	282	31 066
İL YOLLARI Provincial Roads	3 802	26 492	248	632	520	2 202	33 896
TOPLAM Total	22 950	40 183	314	668	520	2 484	67 119



- 1st Division - İstanbul
- 2nd Division - İzmir
- 3rd Division - Konya
- 4th Division - Ankara
- 5th Division - Mersin
- 6th Division - Kayseri
- 7th Division - Samsun
- 8th Division - Elazığ
- 9th Division - Diyarbakır
- 10th Division - Trabzon
- 11th Division - Van
- 12th Division - Erzurum
- 13th Division - Antalya
- 14th Division - Bursa
- 15th Division - Kastamonu
- 16th Division - Sivas
- 18th Division - Kars


K A R A D E N İ Z
M A R M A R A D E N İ Z İ
E G E D E N İ Z İ
A K D E N İ Z
KUZEY KIBRIS TÜRK CUMHURİYETİ
K.K.T.C. GÜZELYURT
LEFKOŞA
BAZMAĞUSA
LARNAKA
GÜNEY KIBRIS RUM YÖNETİMİ
TUZ GÖLÜ
VAN GÖLÜ
İŞARETLER
LEGEND
YER BULMA CETVELİ
LOCATION INDEX
KARAYOLLARI GENEL MÜDÜRLÜĞÜ BÖLGE VE ŞUBE MERKEZLERİ
GENERAL DIRECTORATE OF HIGHWAYS DISTRICT AND SECTION CENTERS

Transportation Statistics

International Network of Road Transport in Turkey





HİZMETE AÇIK OTOYOLLAR

YİD MODELİ İLE YAPIMI DEVAM EDEN OTOYOLLAR

YİD MODELİ İLE HİZMETE AÇILAN OTOYOLLAR

YİD (YAP İŞLET DEVRET)


KARADENİZ
BLACK SEA
AKDENİZ
MEDITERRANEAN SEA
BG
GR
GE
AZ
AR
IR
IRQ
NA
TR
Kapıkule
Aziziye
KIRKLARELİ
EDİRNE
İpsala
TEKİRDAĞ
Gelibolu
Lapseki
Eceabat
ÇANAKKALE
İSTANBUL
İZMİT
ADAPAZARI
BOLU
Gerede
KASTAMONU
SAMSUN
Merzifon
AMASYA
TOKAT
ORDU
GİRESUN
Poti
BATUMI
Sarp
Vale
Türkgözü
Posof
KARS
RİZE
TRABZON
GÜMÜŞHANE
ERZURUM
Refahiye
Horasan
Aşkale
ERZİNCAN
AĞRI
IĞDIR
Dilucu
Doğubeyazıt
Gürbulak
BURSA
BALIKESİR
ESKİŞEHİR
Sivrihisar
ANKARA
YOZGAT
SİVAS
MANİSA
İZMİR
Çeşme
UŞAK
AFYON
AKSARAY
NEVŞEHİR
NİĞDE
KONYA
AYDIN
DENİZLİ
ANTALYA
Pozantı
Toprakkale
MERSİN
ADANA
GAZİANTEP
ŞANLIURFA
MARDİN
Habur
DİYARBAKIR
BİTLİS
VAN
HATAY
Cilvegözü
Yayladağı
E-70
E-80
E-84
E-87
E-88
E-89
E-90
E-91
E-95
E-96
E-97
E-98
E-99
E-691
E-881
E-981
E-982
E YOLLARI
E ROADS IN TURKEY
E-70 529 km
E-80 1 803 km
E-84 151 km
E-87 1 086 km
E-88 587 km
E-89 125 km
E-90 1 930 km
E-91 180 km
E-95 92 km
E-96 439 km
E-97 249 km
E-98 56 km
E-99 808 km
E-691 277 km
E-881 570 km
E-981 403 km
E-982 68 km
TOPLAM: 9 353 Km
(TOTAL)
2016

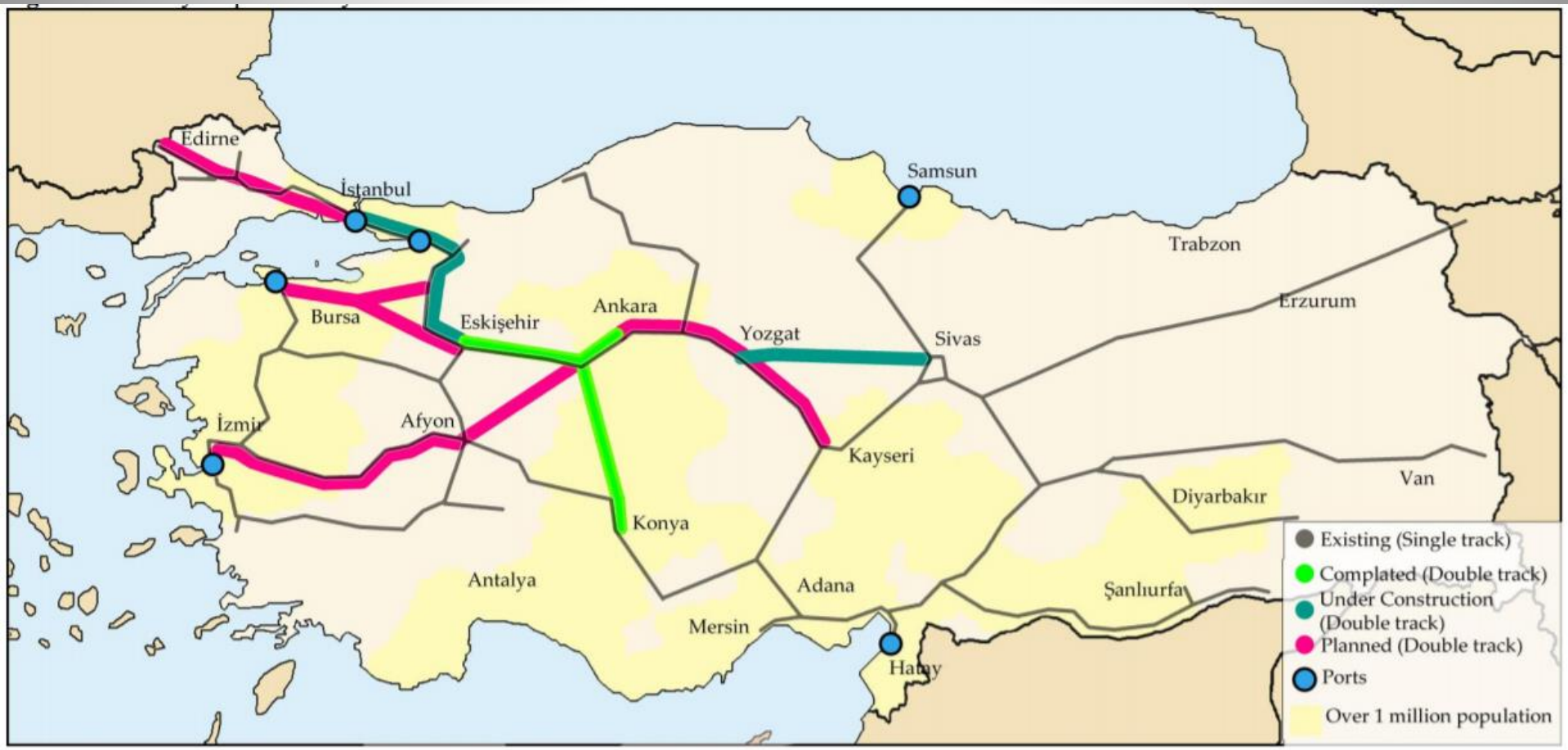


TÜRKİYE'DEKİ ULUSLARARASI YOL GÜZERGAHLARI
INTERNATIONAL ROADS IN TURKEY
TOTAL (KM)

TEM	Trans European North South Motorway	6.940
E Yolları	E Routes	9.553
KEİ Yolları	BSEC Routes	4.472
İEİ Yolları	ECO Routes	9.914
ESCAP Yolları	ESCAP Routes	5.268
Avs.Kat.As.Ulaş.Kor.	TRACECA	11.582
Avrasya Ulaş. Bağl.	BAIL	5.063

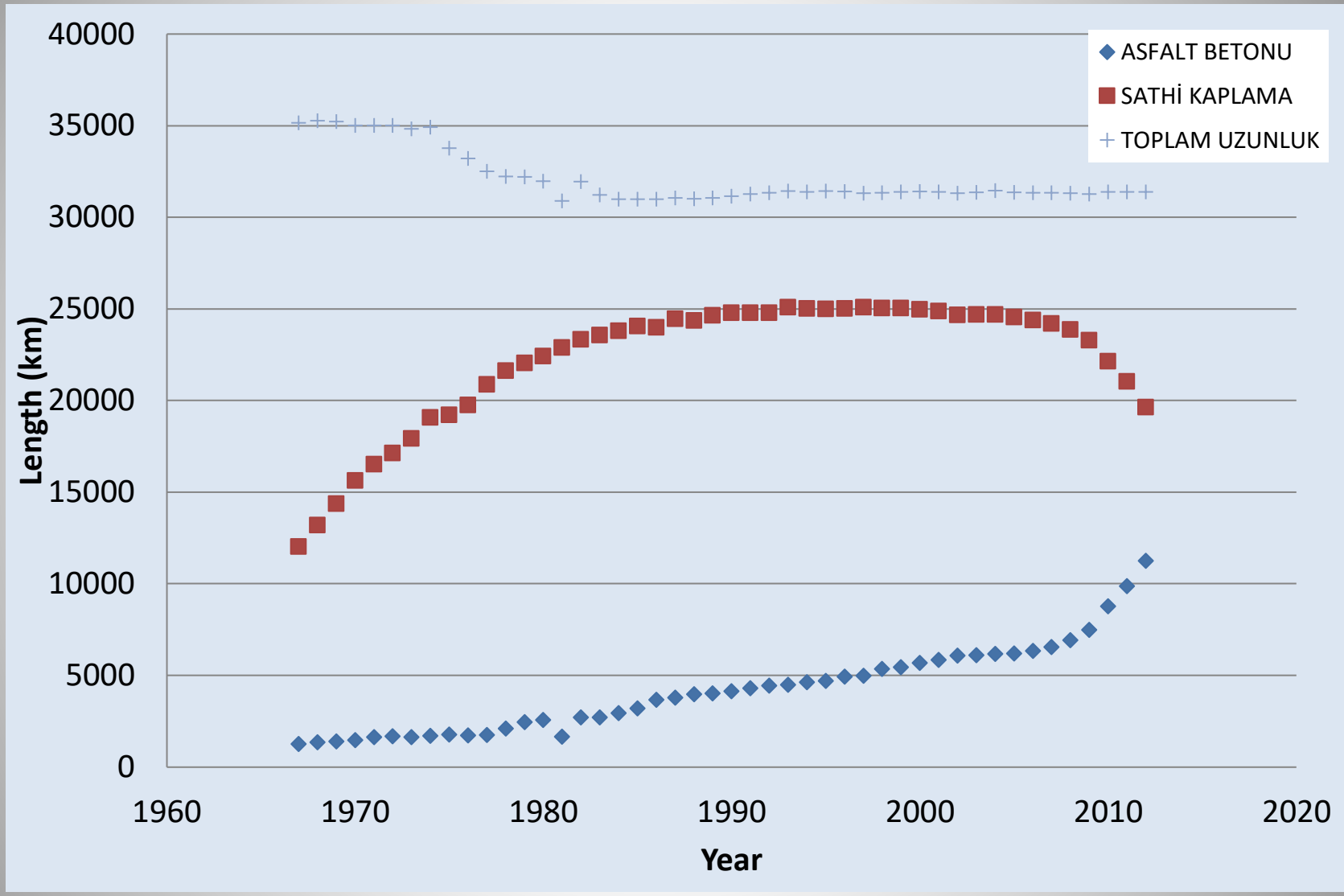
Transportation Statistics

International Network of Rail Transport in Turkey



Road Centerline Length (km/1000 mm^2)

Country	Railway (Km)	Highway (Km) *
Suriye	9.6	224.1
Türkiye	11.2	454.5
Yunanistan	18.3	886.4
Portekiz	30.6	786.1
Danimarka	52.7	1661.3
Fransa	53.5	1794.9
İtalya	54.1	1592.6
Hollanda	67.7	3031.3
Almanya	101.1	1803.9
Lüksemburg	105.8	2000
Belçika	115.4	4859
Çek Cum.	118.7	703.4


40000
35000
30000
25000
20000
15000
10000
5000
0
Length (km)
1960
1970
1980
1990
2000
2010
2020
Year
ASFALT BETONU
SATHİ KAPLAMA
TOPLAM UZUNLUK

Transportation Statistics

Modal Distribution of Freight Movement in Some Countries

USA

%27 Hihgway,
%38 Railway,
%24 Sea
%10 Airway

Germany

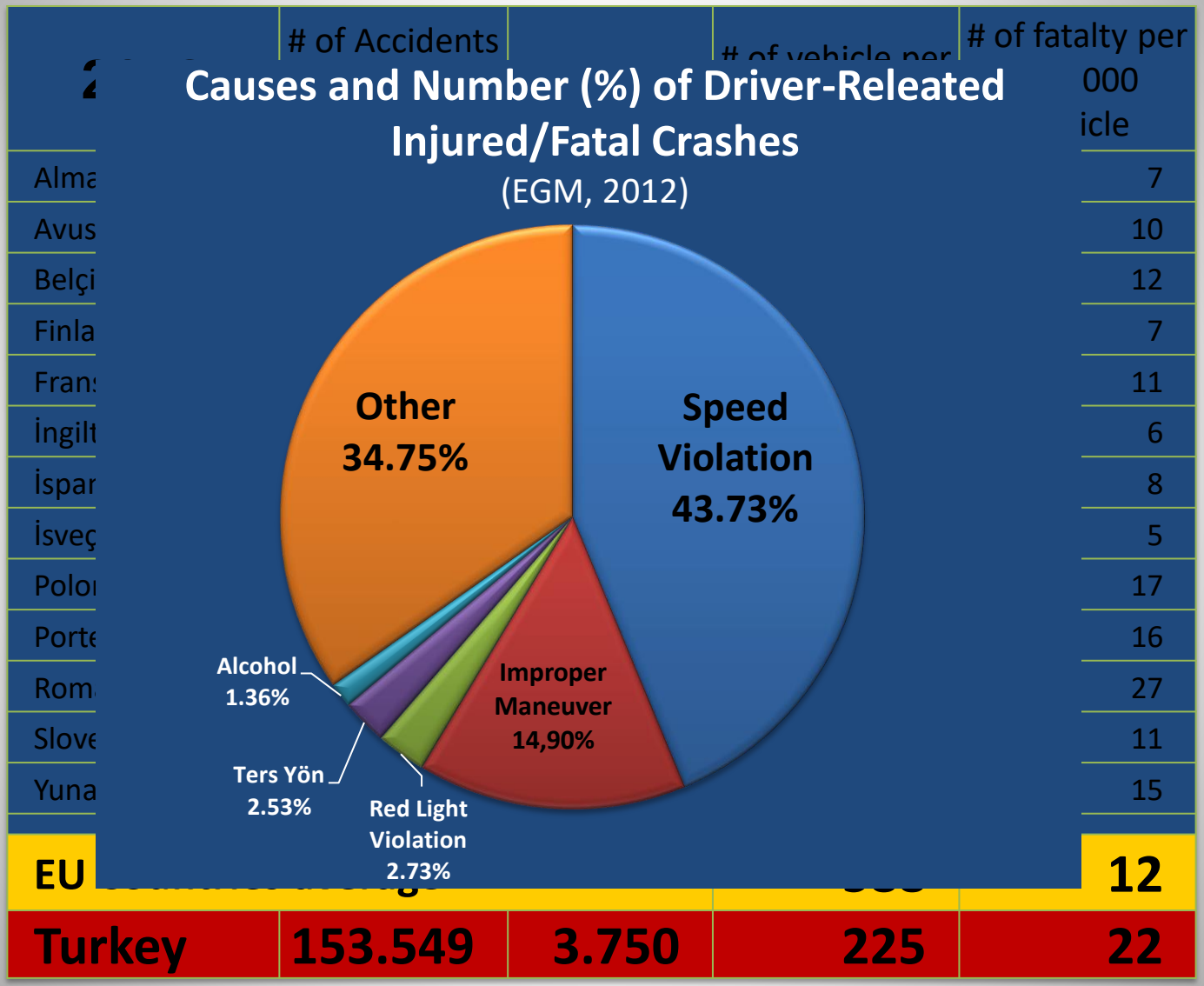
%58 Highway,
%22 Railway,
%12 Sea
%7 Airway

Turkey

% 95 Highway,
% 3 Railway,
% 0,1 Sea
% 2 Airway

Traffic Accident Statistics

Traffic Accidents TUİK 2012



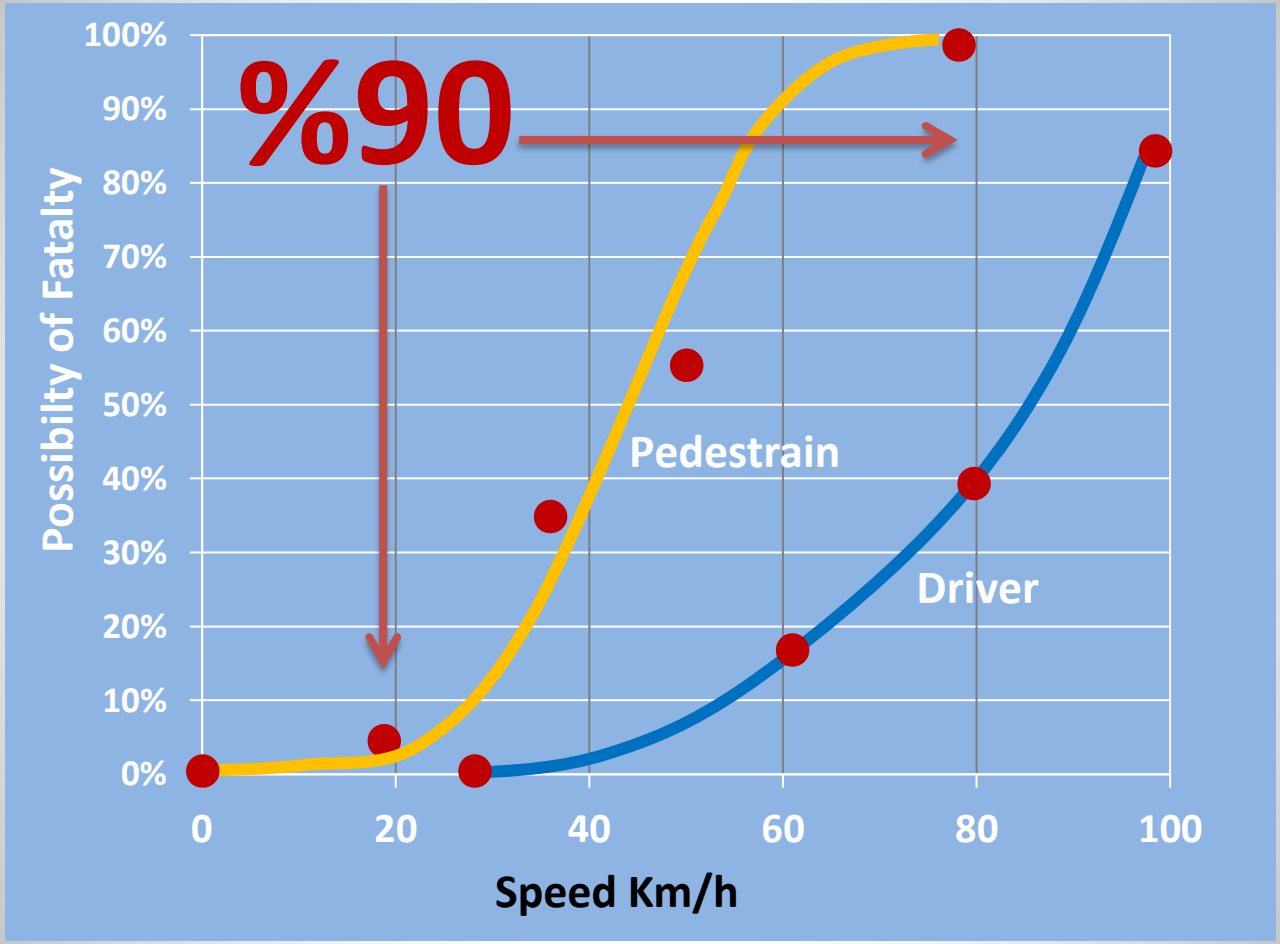
	# of Accidents		# of vehicle per	# of fatalty per 000 icle
Alma				7
Avus				10
Belçi				12
Finla				7
Fran				11
İngilt				6
İspar				8
İsveç				5
Polo				17
Porte				16
Rom				27
Slove				11
Yuna				15
EU				**12**
Turkey	**153.549**	**3.750**	**225**	**22**

%71 of Accident Cause by Speed Violation



Driver Violation, 2012	
Driver Vioaltion	Rate (%)
51/2-a	**1,35**
51/2-b	**0,99**
52/1-a	**33,80**
52/1-b	**34,69**
52/1-c	**0,27**
Other	28,91
Total	**100**

Why Speeding is a risk? *



** Pasanen E, 1991 ve Joksch, 1993*

# of Speed Violation Ticket per 1.000 driver (annual)			
Coutnry	**2006**	**2007**	**2008**
Holland	543	595	558
Austria	327	458	456
Sweden	350	335	n/a
France	114	127	138
Norvey	52	52	51
Turkey*			**18**

The European Transport Safety Council

Referance: (ETSC 2010, 4th PIN Report)

* Emniyet Genel Müdürlüğü (2010 istatistiği)

What is ITS?

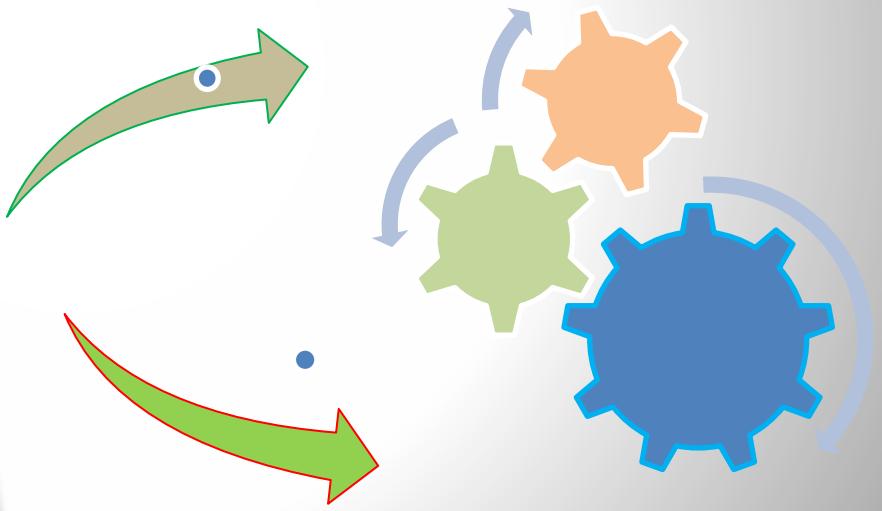
- The application of information and telecommunications technologies to the management and operation of transportation systems
- Electronics, communications, and information processing used in combination to improve the performance of transportation systems.

https://www.youtube.com/watch?v=7icGIHPOviQ

ITS Concepts

- Gather, use, and distribute information
- Integration
- Multi-disciplinary



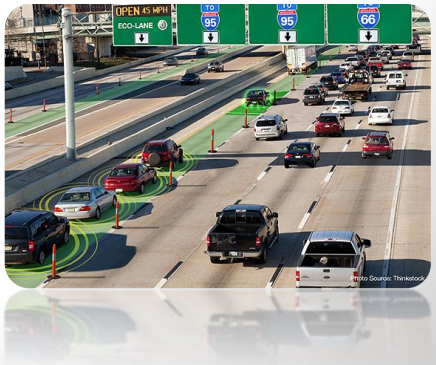


ITS versus Conventional Improvement

- Cost
- Adverse Impacts
  - more capacity-→ more demand
- Right-of-way
- Not all transportation problems can be solved by conventional improvements









ITS Fuctional Areas

- ATMS - Advanced Traffic Management Systems
- ETC - Electronic Toll Collection
- ATIS - Advanced Traveler Information Systems
- APTS - Advanced Public Transportation Systems
- EMS - Emergency Management System
- CVO - Commercial Vehicle Operation
- AVCS- Advanced Vehicle Control Systems